

Kiarash Farzad

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SUMMARY

Sustainability-driven engineer with a strong background in numerical modeling and data-driven analysis. Proven ability to handle large-scale datasets and complex simulations, ranging from atmospheric transport models to building energy systems. Experience includes collaborative projects with national agencies (EPA, NOAA) and the practical application of ASHRAE standards to support LEED and Passive House compliance.

EDUCATION

Northeastern University , Boston, MA	December 2025
PhD in Interdisciplinary Engineering	
Amirkabir University of Technology (Tehran Polytechnic) , Tehran	February 2020
MSc in Civil and Environmental Engineering	
Islamic Azad University Central Tehran Branch , Tehran	June 2017
BSc in Civil Engineering	

EXPERIENCE

R.W. Sullivan , Boston, MA	October 2025 – Present
Building Energy Modeler Intern	
- Simulate and analyze building energy consumption using TRACE™ 700 and WUFI® Passive for VRF, DOAS, ERV, and WSHP systems. Define thermal zoning and apply ASHRAE 90.1 standards to calculate window shading and wall derating factors, and provide alternative scenarios for clients. - Conduct BERDO emissions analysis across multiple energy model scenarios to identify the best alternative scenario for regulatory compliance and cost reduction for the client. - Modify project geometry in response to Phius review to meet Passive House requirements, execute exterior lighting calculations to maximize credit savings for LEED Minimum Energy Performance.	

RESEARCH EXPERIENCE

Northeastern University , Boston, MA	September 2021 – December 2025
Doctoral Researcher	
- Developed a statistically downscaled Chemical Transport Model (CTM) product and a custom health impact assessment tool that decreased forecast errors by 20%, providing a daily US-wide 1-km air quality dataset by leveraging a satellite dataset with 21 million grid cells. - Peer-Reviewed Manuscript: Farzad, K., et al. Sci. Total Environ., 2025 - Simulated air quality using CTMs on a high-performance computing (HPC) cluster and validated spatiotemporal results through statistical, spatial, and time-series analyses of multi-platform observational datasets (e.g., satellite), integrating data from SQL databases and NetCDF files. - Enhanced the NOAA National Air Quality Forecast Capability (NAQFC) system by upgrading the core CTM (CMAQ), leading to a 2% increase in PM_{2.5} forecasting accuracy. - Conducted air pollution source apportionment across Greater Boston using CAMx PSAT, identifying major emission sources and regions to offer actionable insights for regional control strategies.	

SKILLS

Programming: R, Python (NumPy, pandas, scikit-learn, TensorFlow), Bash, SQL, MATLAB, Fortran, csh

Energy Modeling & Software: WUFI®, TRACE™ 700, SketchUp, Bluebeam, AutoCAD, FreeCAD

Atmospheric Modeling & Software: CMAQ, CAMx, WRF, BenMAP, QGIS, ArcGIS, AERMOD, GCAM/GLIMPSE

Data Science: Time Series Analysis, Machine Learning (GAM, RNN/LSTM), Statistical Analysis, High-Performance Computing (HPC), Parallel Computing, Git